

Modeling and Calibrating: LLMs Figure out their Decoding Process

Matthew Khoriaty

Project write-up, *Reliable ML with Unreliable Black Boxes*

Abstract

An autoregressive language model is trained to predict text, so when it generates text it has an incentive to model its own generation process — including the decoder that samples from its logits. We accumulate evidence that models do exactly this: they infer features of their own decoder (a token blacklist, a top- k truncation, a temperature) from the text in their context, and shift their predictions accordingly. We build a streaming Bayesian filter that estimates the model’s inferred inverse-temperature $\hat{\beta}_t$ online and a corrector that divides a user’s target by $\hat{\beta}_t$. An empirical sweep shows the corrector pins the effective temperature in the “boredom” regime ($\beta^{\text{target}} > 1$) but overshoots in the “confusion” regime ($\beta^{\text{target}} < 1$). Probing what the filter measures sharpens the picture and revises the original framing: on real text the filter is *not* an online version of Guo temperature scaling (it sits ~ 1 nat in $\log \hat{\beta}$ away from the post-hoc MLE, e.g. -0.937 vs $+0.000$ on WikiText), and Qwen-2.5-3B is already well calibrated on naturalistic English. A controlled sampler-switch experiment falsifies the strong hypothesis that the belief is a fast posterior read off per-token entropy. The belief that survives is slower and more structural — carried by rank statistics and blacklist signatures, not instantaneous entropy. Finally, two metrics for the corrector disagree: closed-loop stability prefers full correction, while invariance on held-out natural text shows full correction is catastrophic unless the filter is given a fading memory ($\gamma < 1$). All numbers and tables in this document are generated from the experiments’ data files by build scripts; the prose cites them by name.

1 Introduction

An autoregressive language model emits, at each step, a probability distribution over next tokens given the prefix so far. *Decoding* is the choice of how to sample from that distribution: temperature scaling, top- k , and top- p all act by sharpening or flattening the model’s distribution before drawing the next token. This project starts from a question about what a language model “thinks” about its own decoding process. Because models are trained for text prediction, not generation, they have an incentive to model their own generation process. We hypothesize, and have begun to validate, that *a language model forms a belief about its own decoding parameters by conditioning on the text it has generated so far*. If you sample at a low temperature, the model notices that the recent text is low-entropy and adjusts its output probabilities to match; the low-temperature samples then further condition the model to expect low-temperature text. This is a feedback loop.

Mirostat (Basu et al., 2021) documented a version of this effect and patched it with a control loop on aggregate per-token surprise. That fix cannot adapt to context, because high- or low-entropy text is sometimes exactly what a response calls for. The broader and (to us) more interesting question is to *model* what an LLM believes about its own decoder. This is close to the calibration literature — post-hoc temperature scaling (Guo et al., 2017), its use on language-model surprisal (Liu et al., 2024), and trained universal calibrators (Shen et al., 2024) — and to the algorithms-with-predictions

viewpoint (Mitzenmacher and Vassilvitskii, 2020) of the course, in which a possibly-unreliable prediction (here, the model’s inferred decoder) augments a downstream decision. It also connects to recent work on online and high-dimensional calibration (Peng, 2025; Fishelson et al., 2025).

Contributions and revisions. The original proposal framed the streaming estimator as “online temperature scaling converging in the large-data limit to the same global T that Guo’s batch MLE fits.” This write-up reports the experiments we ran to test that framing, and the framing does not fully survive. We find: (i) positive, model-internal evidence that the model infers its decoder (§3); (ii) a corrector that works asymmetrically (§4); (iii) that the filter is *not* Guo MLE on real text and that the well-calibrated-model regime leaves no post-hoc temperature gain (§5); (iv) that the belief is not a fast entropy read-out (§6); and (v) that two natural criteria for tuning the corrector disagree, with memory decay as the load-bearing parameter (§7). The belief-imprint hypothesis is *refined and constrained*, not refuted.

2 Method: a streaming Laplace filter on $\log \beta$

Let $\ell_t \in \mathbb{R}^V$ be the model’s pre-decoding logits at step t and x_t the token actually emitted. We posit a matched-prior *imprint model*: the model behaves as if the natural-reference logits are $\ell_t^* = \ell_t / \hat{\beta}_t$, i.e. the observed logits are the reference scaled by the model’s current belief $\hat{\beta}_t$ about its own inverse temperature. Placing a log-normal prior on β (mean 1, scale σ_0) and doing a per-token Laplace (Newton) update of the posterior on $\eta = \log \beta$ gives a streaming estimator whose update uses the score and Fisher information of the categorical likelihood under $q = \text{softmax}(\ell_t)$:

$$\text{score}_t = \ell_t[x_t] - \mathbb{E}_q[\ell_t], \quad J_t = \frac{1}{\sigma_0^2} + \sum_{\tau \leq t} \text{Var}_q[\ell_\tau].$$

Crucially q , $\mathbb{E}_q[\ell_t]$, and $\text{Var}_q[\ell_t]$ are evaluated at $\beta = 1$ and never re-evaluated at the running $\hat{\beta}_t$: the imprint model is defined precisely so the score and Fisher are $\hat{\beta}$ -independent. This makes the filter a per-token Bayesian update under the imprint model, *not* an iterated Newton solver on a global likelihood — a distinction that becomes important in §5. Three knobs parameterize the belief dynamics: an evidence weight α (per-step step size), a memory decay γ (exponential forgetting; $\gamma = 1$ is no decay, $\gamma < 1$ is a fading window of effective length $1/(1 - \gamma)$), and a correction exponent p .

If the imprint is a single scalar rescaling, dividing the user’s target by the estimate yields a corrector that pins the effective sampling distribution to target:

$$\beta_t^{\text{dec}} = \beta^{\text{target}} / \hat{\beta}_t^p, \quad p = 1 \text{ (full)}, \quad p = 0 \text{ (off)}, \quad p \in (0, 1) \text{ (partial)}.$$

Everything below uses Qwen-2.5-3B (base, fp16). The filter implementation and its unit tests (shift-invariance, batched = serial equivalence, recovery of β_{true} under matched-prior synthetic data) live in `src/decoding_decoding/counter_decode.py`.

3 The model infers its decoder: positive evidence

Before modeling the belief, we establish that there is one. Two controlled experiments isolate the model’s response to a known decoder manipulation, holding content fixed and comparing treatment to control with cluster-bootstrap confidence intervals.

Blacklist self-prediction. When the k highest-probability tokens are banned (set to zero probability) at sampling time, the model’s own softmax probability on the banned token drops between early and late sequence positions — it learns its blacklist from the absent-token signature. For the rank-1 ban (the token ‘the’), the late-minus-early within-treatment change is $\Delta p_{\text{late-early}} = -0.019$, and the paired late treatment-vs-control difference is -0.016 with 95% CI $[-0.019, -0.013]$, clear of zero (Table 1, Fig. 1).

Table 1: Blacklist self-prediction (run 2026-05-10). CI is the cluster-bootstrap 95% interval on the late treatment-minus-control difference.

banned rank	token	$\Delta p_{\text{late-early}}$	treat.-ctrl (late)	95% CI
1	the	-0.019	-0.016	$[-0.019, -0.013]$
2	,	-0.017	-0.019	$[-0.022, -0.016]$
3	.	-0.017	-0.018	$[-0.021, -0.015]$
4	of	-0.011	-0.004	$[-0.005, -0.002]$
5	to	-0.005	-0.000	$[-0.002, 0.001]$

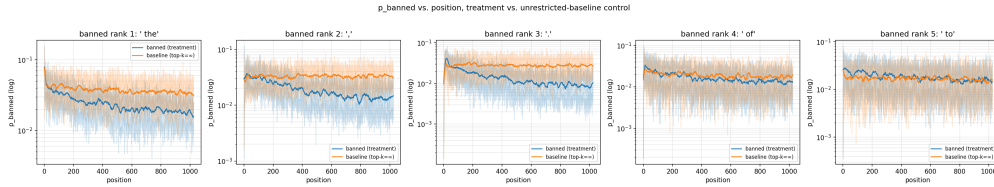


Figure 1: Model probability assigned to the banned token vs. sequence position, per banned rank. Probability mass on a commonly-banned token decays as the model infers the ban from its absence in the context.

Top- k self-prediction. Under top- k sampling, the model’s predicted-rank distribution develops a cliff at rank K that exceeds the corresponding cliff under unrestricted sampling. The treatment-minus-control gap at rank K is $+0.475$ for $k = 3$ and $+0.170$ for $k = 5$ (Table 2). Within-treatment concentration grows between early and late positions for finite k and *shrinks* for $k = \infty$ — this is not generic drift but a response to the truncation.

Table 2: Top- k self-prediction ($L = 2048$, run 2026-05-09): the extra cliff at rank K in the predicted-rank distribution, treatment minus control.

k	treat.-ctrl at rank K	95% CI
3	$+0.475$	$[0.395, 0.567]$
5	$+0.170$	$[0.150, 0.191]$

These two results are the load-bearing positive evidence: the model demonstrably conditions its predictions on which tokens its decoder did and did not allow.

4 The corrector and concentration dynamics

We next put the filter in a closed loop. Self-generating 200 tokens per prompt at six values of β^{target} , we compare an *uncorrected* arm (fixed $\beta^{\text{dec}} = \beta^{\text{target}}$) against a *corrected* arm ($\beta_t^{\text{dec}} = \beta^{\text{target}} / \hat{\beta}_t$).

Concentration drift. The uncorrected arm sharpens for $\beta^{\text{target}} > 1$ (a “boredom trap”) and flattens for $\beta^{\text{target}} < 1$ (a “confusion trap”); the corrected arm keeps the sampling distribution bounded in both directions (Table 3, Fig. 2). The $\beta^{\text{target}} = 1$ row is a negative control, where the two arms coincide.

Table 3: F0 concentration, late trajectory ($t \in [150, 200]$), corrected vs. uncorrected: max-probability and entropy of the sampling distribution.

β^{target}	corr. max P	unco. max P	corr. H	unco. H
0.5	0.579	0.001	1.94	11.47
0.7	0.521	0.012	2.29	10.48
1.0	0.536	0.534	2.17	2.21
1.3	0.545	0.755	2.11	0.88
1.7	0.602	0.856	1.76	0.44
2.0	0.603	0.891	1.70	0.32

An asymmetry. The corrector is not symmetric. In the boredom regime it pins concentration near its $t = 0$ value, as the theory predicts. In the confusion regime it only partially works: the corrected late max- P_{sample} at $\beta^{\text{target}} = 0.5$ settles at 0.579, well above where it starts — an upward drift the strict (flat-in- t) prediction does not allow. The mechanism is visible in the filter: under uncorrected $\beta^{\text{dec}} = 0.5$, the streaming estimate runs to $\log \hat{\beta} = -4.74$ by $t = 200$ (Table 4), an overshoot. Dividing the target by an overshooting $\hat{\beta}$ produces an over-large β^{dec} , hence the residual sharpening.

Table 4: Streaming $\log \hat{\beta}$ at $t = 200$ under uncorrected decoding (F1.2): sharp decoders push it up, flat decoders push it down, asymmetrically.

β^{dec}	median $\log \hat{\beta}_{t=200}$	IQR
0.5	-4.74	1.25
0.7	-2.71	0.78
1.0	-0.05	0.48
1.3	+0.67	0.35
1.7	+0.98	0.27
2.0	+1.08	0.23

Closed-loop check and multiplicativity. The corrected arm pins $\beta^{\text{eff}} = \beta^{\text{target}}$ by construction (Table 5: the corrected error $E(T)$ is numerically zero), so this is a tautological check that the loop ran; the uncorrected error grows worst at $\beta^{\text{target}} = 0.5$, where $E(T) = 4.74$ nats (a drift of more than two orders of magnitude in β^{eff}). The imprint is *approximately* multiplicative: an SVD of the centered across-decoder logit matrix leaves an off-axis residual fraction that grows to $R = 0.15$ by $t = 200$, so 15% of the across-decoder variation is not explained by a single scalar β .

5 What does the filter actually measure?

The corrector’s confusion-regime failure motivates asking what the filter’s $\hat{\beta}$ is. We test the proposal’s claim — that it is online Guo temperature scaling — directly, and find it false on real

Table 5: Closed-loop error $E(T) = \text{median} |\log \beta^{\text{eff}} - \log \beta^{\text{target}}|$ at $t = 200$ (F3).

β^{target}	$E(T)$ corrected	$E(T)$ uncorrected
0.5	$1.9e-9$	4.74
0.7	$4.5e-9$	2.71
1.0	$0.0e+0$	0.25
1.3	$5.6e-8$	0.67
1.7	$1.3e-8$	0.98
2.0	$1.9e-9$	1.08

text.

No filter pathology under self-sampling. First we rule out the weakest alternative: that the filter drifts on its own. Feeding the filter i.i.d. samples from $\text{softmax}(\ell_t)$ at $\beta^{\text{dec}} = 1$ (the no-imprint null), $\log \hat{\beta}$ flatlines — median -0.070 on WikiText self-generation and -0.052 on WritingPrompts, both within one standard error of zero (Fig. 3). On real Wikipedia continuations the filter agrees with self-sampling (paired $t = -0.52$), but on real WritingPrompts it drifts negative: a paired difference of -0.304 in $\log \beta$ ($t = -3.80$, $p < 10^{-3}$). So the model is well calibrated on Wikipedia and slightly over-confident on Reddit creative writing — a real out-of-domain calibration offset, but an order of magnitude smaller than the drift seen under extreme β -decoding. The closed-loop drift is not a filter artifact.

The streaming filter is not Guo MLE (Exp A). On a proper per-corpus calibration split, the Guo MLE sits essentially at $\log \beta = 0$ on both corpora ($+0.000$ on WikiText, -0.036 on WritingPrompts), confirming Qwen-2.5-3B is already calibrated. The pooled streaming filter on the *same* tokens settles at $\log \hat{\beta} = -0.937$ (WikiText) and -0.802 (WritingPrompts) — about a nat away (Table 6, Fig. 4). The two estimators have *different fixed points*: the filter is a single damped Newton step from the prior mode under the imprint model, accurate near $\beta = 1$ but biased as it walks up the flat NLL surface, whereas the Guo MLE is the global minimizer. Applied as a calibrator the filter *hurts*: on WritingPrompts test, the per-passage streaming $\hat{\beta}$ raises per-token PPL from 19.553 to 24.989 ($\Delta \log \text{PPL} = +0.2522$, $t = +26.46$), while the Guo MLE is statistically indistinguishable from $T = 1$ (paired $t = -0.95$, $p = 0.347$). The proposal’s asymptotic-equivalence claim does not survive; “calibration” and “imprint detection” are distinct targets.

Table 6: Calibration $\hat{\beta}$ per estimator (as $\log \hat{\beta}$; $0 \Leftrightarrow \hat{\beta} = 1$). The pooled streaming filter and the Guo MLE disagree by ≈ 1 nat.

corpus	$\log \hat{\beta}_{\text{stream_med}}$	$\log \hat{\beta}_{\text{stream_pooled}}$	$\log \hat{\beta}_{\text{MLE}}$
WikiText	-0.059	-0.937	+0.000
WritingPrompts	-0.314	-0.802	-0.036

Does the sliding filter track genre (Exp B)? With the calibration framing gone, the remaining question is whether a sliding $\hat{\beta}_t$ tracks content transitions at all. On a sequence alternating WikiText and WritingPrompts chunks, a fading-memory filter ($\gamma = 0.90\text{--}0.95$) sees a small genre signal — the within-chunk mean of $\log \hat{\beta}$ is $+0.054$ higher on WikiText than WritingPrompts, the

right sign — but it is dwarfed by the 1.34-nat difference in pre-decoding entropy between the genres (Table 7, Fig. 5). With no decay ($\gamma = 1$) the filter saturates to its pooled-streaming bias ($\log \hat{\beta} \approx -0.46$) and loses all per-context sensitivity. The signal is plausibly content-driven, not a clean read of the decoder.

Table 7: Exp B: mean $\log \hat{\beta}$ per genre (boundary-stripped chunk interiors) and mean pre-decoding entropy per genre, by memory decay γ .

γ	$\log \hat{\beta}$ wiki	$\log \hat{\beta}$ WP	WT-WP	H wiki	H WP
0.90	-0.084	-0.138	+0.054	1.686	3.024
0.95	-0.091	-0.142	+0.051	1.686	3.024
0.98	-0.115	-0.143	+0.028	1.686	3.024
0.99	-0.142	-0.140	-0.002	1.686	3.024
1.00	-0.458	-0.463	+0.004	1.686	3.024

6 The form of the belief: not a fast entropy read-out

If the model has a belief about β^{dec} , does it show up as pre-decoding entropy that snaps to a new sampler within a few dozen tokens? We self-generate 150 tokens at three β^{dec} and read the pre-decoding entropy $H(\text{softmax}(\ell_t))$ in a late window. The raw effect is enormous: late-window H moves from 1.172 nats at $\beta^{\text{dec}} = 2.0$ to 9.216 nats at $\beta^{\text{dec}} = 0.5$ — a paired spread of 8.04 nats ($t = +69.7$, all 32 prompts ordering the same way; Table 8, Fig. 6).

Table 8: Exp C: late-window ($t \in [80, 150)$) pre-decoding entropy by β^{dec} , with per-prompt paired difference vs. $\beta = 2.0$.

condition	late-window H (nats)	paired diff vs $\beta = 2.0$
$\beta_{\text{dec}} = 0.5$	9.216 ± 0.052	$+8.045 \pm 0.115$ ($t = +69.7$, $n_+ = 32/32$)
$\beta_{\text{dec}} = 1.0$	2.168 ± 0.186	$+0.997$
$\beta_{\text{dec}} = 2.0$	1.172 ± 0.109	—

But it is garbage-in, garbage-out — and the swap control proves it. Natural high-entropy human writing sits far below 9 nats, and the $\beta = 0.5$ generations are token salad within ~ 20 steps: the entropy climbs because each random token makes the context less parseable, not (necessarily) because the model believes the decoder is hot. To separate decoder-belief from content quality, we hold the prefix’s regime fixed and *switch* the sampler at step 80. If the belief were a fast posterior on recent tokens, switching $0.5 \rightarrow 1.0$ should pull late entropy back to the $\beta = 1$ baseline within a handful of tokens. It does not: the `pre05.post10` condition stays essentially flat, from 8.56 (pre-switch) to 8.91 (50 tokens after the switch; Table 9, Fig. 7). Switching $0.5 \rightarrow 2.0$ *does* recover coherence, but only because $\beta = 2$ forces near-top-1 selection that re-anchors the context to in-distribution English — the sampler overwriting the bad prefix, not the model updating a belief.

What this constrains. The decisive comparison falsifies the strong instantiation in which decoder-belief is a fast entropy read-out. It does *not* refute the broader claim — the blacklist and top- k experiments (§3) give direct positive evidence, and the WritingPrompts offset (§5) shows the model

Table 9: Exp C swap control: median pre-decoding H before the switch and at offsets after it. The decisive row is `pre05_post10`, which does not recover.

condition	$\beta_{\text{pre}} \rightarrow \beta_{\text{post}}$	pre-med	+0	+5	+10	+20	+50
<code>pre05_post05</code>	$0.5 \rightarrow 0.5$ (control)	8.42	9.25	9.21	9.49	9.66	9.47
<code>pre05_post10</code>	$0.5 \rightarrow 1.0$	8.56	9.44	9.10	9.10	8.90	8.91
<code>pre05_post20</code>	$0.5 \rightarrow 2.0$	8.57	9.53	6.83	4.33	1.68	0.33
<code>pre20_post05</code>	$2.0 \rightarrow 0.5$	1.21	0.69	5.20	6.46	8.01	9.06
<code>pre20_post20</code>	$2.0 \rightarrow 2.0$ (control)	1.29	0.64	0.58	1.05	0.30	0.39

does respond to which tokens its context contains. The belief operates on slower, more aggregate features — rank statistics, support shape, blacklist signatures — not instantaneous entropy.

7 Belief-update dynamics: closed-loop vs. invariance

Finally we ask which (γ, α, p) best parameterize the belief dynamics, using two complementary criteria. *Closed-loop badness* self-generates under the corrected sampler and penalizes a corrector that fails to track β^{target} . *Belief invariance* teacher-forces the filter on a held-out natural-text continuation after a prefix sampled at varying β^{prefix} , and asks that the corrected per-token cross-entropy be *invariant* to the prefix (a well-calibrated corrector should cancel the prefix imprint and leave the natural distribution untouched).

The two criteria disagree. By closed-loop badness, full correction wins: the default ($\gamma = 1, \alpha = 1, p = 1$) is best (badness 0.011; Table 10). By invariance, full correction at $\gamma = 1$ is catastrophic: cross-entropy on $\beta^{\text{prefix}} = 0.5$ continuations explodes from 3.60 nats at $p = 0$ to 19.44 nats at $p = 1$ (near-uniform over the vocabulary), and the only configurations that stay near the no-correction floor have $\gamma < 1$ (Table 11, Fig. 8).

Table 10: Closed-loop sweep, top 5 by badness. $H@ \beta^{\text{target}}$ and $M@ \beta^{\text{target}}$ are late-window mean entropy and max- P_{sample} .

γ	α	p	$H@0.5$	$H@1$	$H@2$	$M@0.5$	$M@1$	$M@2$	badness
1.00	1.00	1.00	2.45	2.21	1.18	0.53	0.54	0.71	0.011
1.00	1.00	0.75	3.12	1.97	0.92	0.44	0.58	0.75	0.046
1.00	0.50	1.00	3.38	2.27	0.89	0.42	0.52	0.75	0.051
1.00	0.50	0.75	4.56	2.18	0.72	0.31	0.53	0.79	0.089
1.00	1.00	0.50	4.55	2.19	0.66	0.30	0.53	0.80	0.105

Resolving the disagreement. Closed-loop badness is a self-consistency illusion: the filter accumulates a biased $\hat{\beta}$ from the corrected sampler’s own output, then divides the next logits by that same biased $\hat{\beta}$, so the two biases cancel *inside the loop*. “The sampler’s output looks orderly” is necessary but not sufficient for “the filter modeled the belief.” Breaking the loop by teacher-forcing on natural text removes the cancellation, and the bias appears as enormous cross-entropy. Invariance is the more direct criterion, because it grounds the corrector on data the model did not generate. Memory decay $\gamma < 1$ is the load-bearing parameter: it bounds the filter’s worst-case bias by the fading window’s effective sample size, holding the CE variance to ≤ 0.07 nats² across

Table 11: Belief-invariance sweep, top 5 by per-prompt CE variance across β^{prefix} (lower is better). All $p = 0$ configs are identical (correction off) and shown as one “any/any” row.

γ	α	p	CE@ $\beta^{\text{pre}}=0.5$	CE@1	CE@2
<i>any</i>	<i>any</i>	0.00	3.599	3.592	3.631
0.85	0.50	0.25	3.596	3.604	3.639
0.95	0.50	0.25	3.583	3.606	3.632
0.85	1.00	0.25	3.596	3.610	3.643
0.95	1.00	0.25	3.580	3.610	3.631

the entire p range (versus the floor 0.017). The evidence weight α matters little. A reasonable default for off-temperature generation ($\sigma_0 = 0.2$) is moderate decay with partial correction, e.g. ($\gamma = 0.95$, $\alpha = 1.0$, $p = 0.5$); for natural generation ($\beta^{\text{target}} = 1$) no correction is best, since the model is already calibrated.

8 Limitations and conclusion

Limitations. All results use one model (Qwen-2.5-3B), one tokenization, and a small set of corpora; the “Qwen is well-calibrated” finding and the magnitude of the WritingPrompts offset are model- and corpus-specific. The imprint is only approximately multiplicative ($R = 0.15$ off-axis), and no scalar- β correction can cancel non-scalar structure (mode collapse, repetition, vocabulary narrowing); we have not measured how much of the prefix imprint is non-scalar. The invariance test uses only $\beta^{\text{target}} = 1$, the trivially-calibrated case, because there is no natural-text ground truth at other targets. Exp C cannot fully separate decoder-belief from garbage-in/garbage-out, because $\beta = 0.5$ decoheres the prefix faster than any belief update could compensate.

Conclusion. Language models do infer features of their own decoder from context (blacklists, top- k cliffs, an over-confidence offset on out-of-domain text). A streaming Laplace filter that models this as a scalar inverse-temperature belief yields a corrector that works in the boredom regime and overshoots in the confusion regime. The filter is not online Guo temperature scaling — it has a different fixed point and hurts as a calibrator — and the belief it measures is not a fast entropy read-out but something slower and more structural. The right way to tune the corrector is by invariance on held-out natural text, not closed-loop stability, and that criterion makes a fading memory ($\gamma < 1$) essential. The belief-imprint hypothesis is sharpened, not refuted; the open theoretical question is a closed-loop fixed-point analysis of the coupled $(\beta_t^{\text{model}}, \hat{\beta}_t)$ dynamics that would explain the confusion-regime overshoot.

References

- Sourya Basu, Govardana Sachitanandam Ramachandran, Nitish Shirish Keskar, and Lav R. Varshney. Mirostat: A neural text decoding algorithm that directly controls perplexity, 2021. URL <https://arxiv.org/abs/2007.14966>.
- Maxwell Fishelson, Noah Golowich, Mehryar Mohri, and Jon Schneider. High-dimensional calibration from swap regret, 2025. URL <https://arxiv.org/abs/2505.21460>.

- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks, 2017. URL <https://arxiv.org/abs/1706.04599>.
- Tong Liu, Iza Škrjanec, and Vera Demberg. Temperature-scaling surprisal estimates improve fit to human reading times – but does it do so for the "right reasons"?, 2024. URL <https://arxiv.org/abs/2311.09325>.
- Michael Mitzenmacher and Sergei Vassilvitskii. Algorithms with predictions, 2020. URL <https://arxiv.org/abs/2006.09123>.
- Binghui Peng. High dimensional online calibration in polynomial time, 2025. URL <https://arxiv.org/abs/2504.09096>.
- Maohao Shen, Subhro Das, Kristjan Greenewald, Prasanna Sattigeri, Gregory Wornell, and Soumya Ghosh. Thermometer: Towards universal calibration for large language models, 2024. URL <https://arxiv.org/abs/2403.08819>.

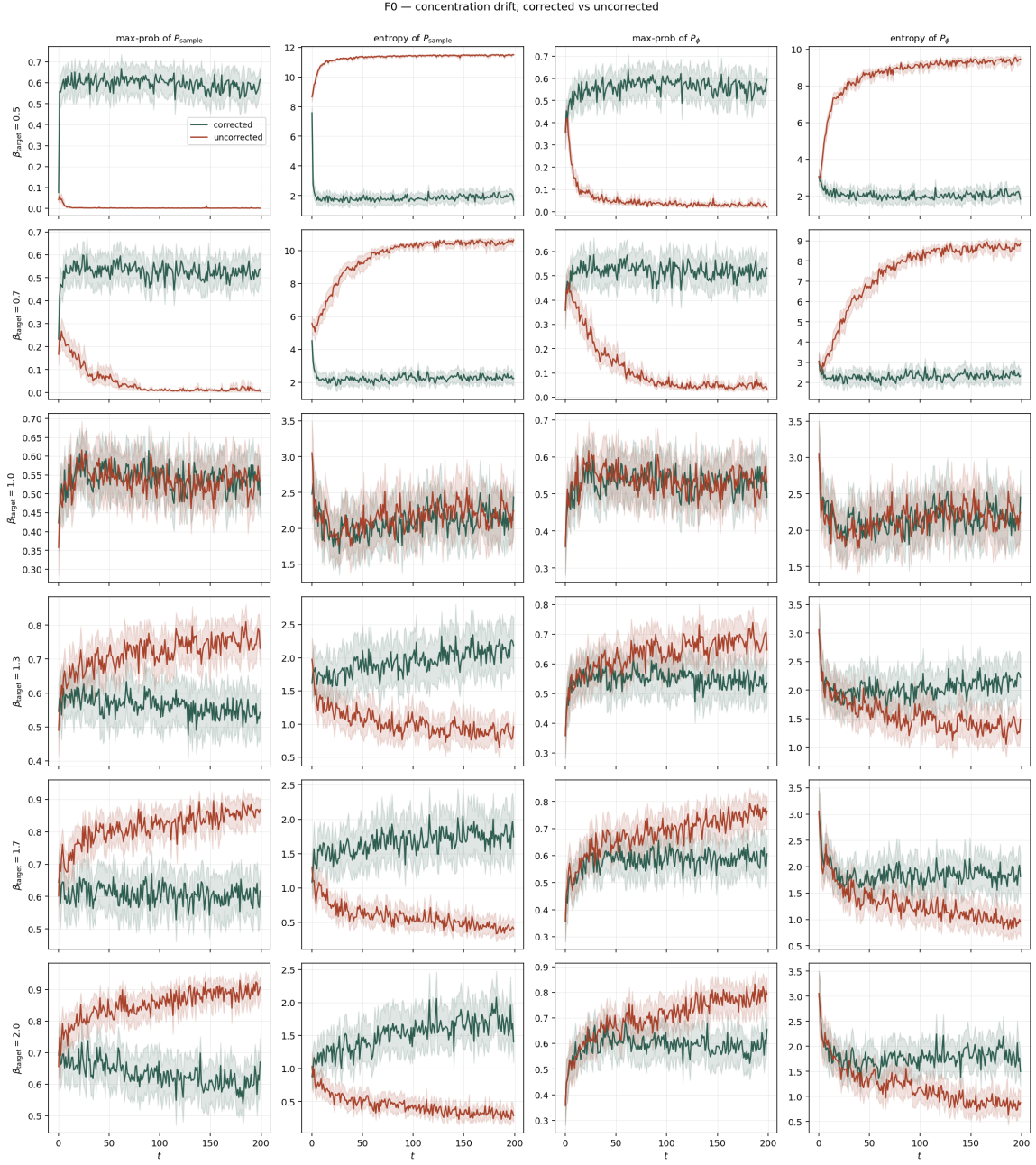


Figure 2: Concentration (max- P and entropy) of the sampling and pre-decoding distributions over t , six β_{target} rows, corrected (green) vs. uncorrected (red). The uncorrected arm runs away in both regimes; the corrected arm stays bounded.

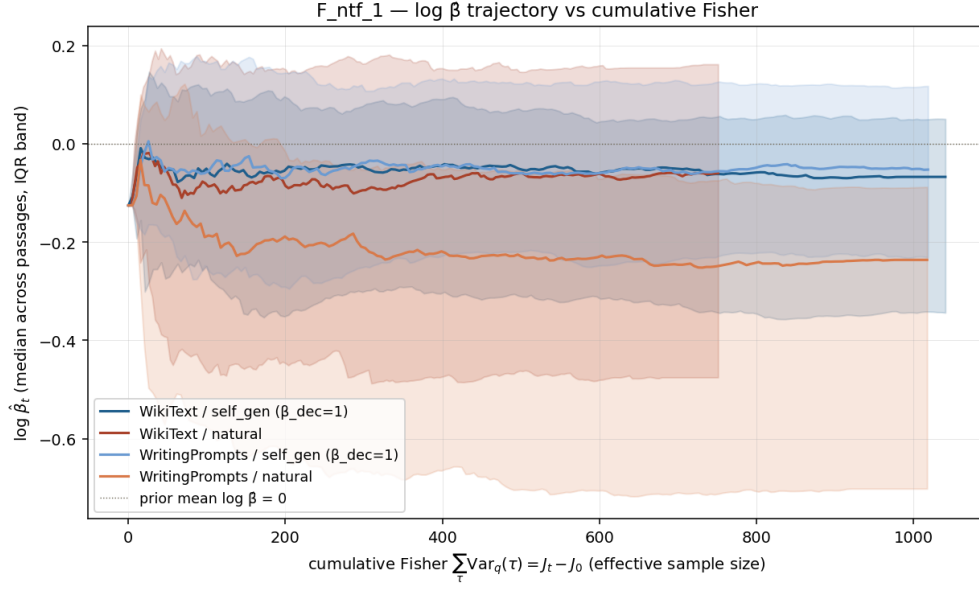


Figure 3: Median $\log \hat{\beta}_t$ vs. cumulative Fisher (the filter’s effective sample size). Self-generation arms flatline at zero; WritingPrompts/natural drifts negative and stays there.

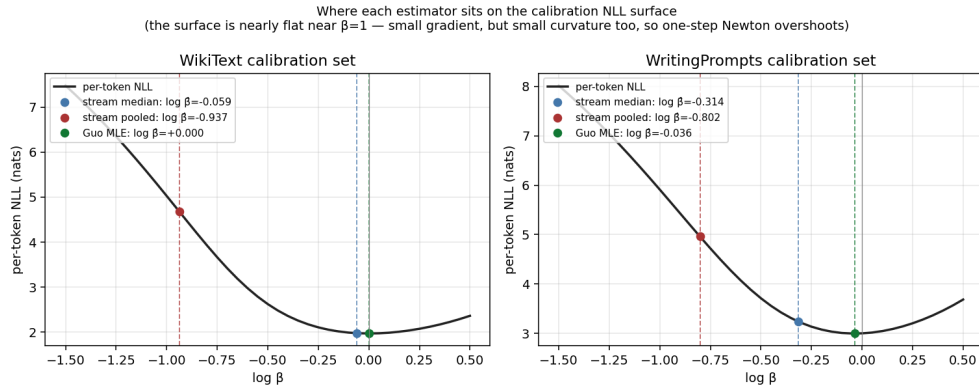


Figure 4: Per-token calibration NLL vs. $\log \beta$. The minimum is at $\log \beta \approx 0$ (Qwen is calibrated); the pooled streaming filter lands far up the steep wall, not at the minimum.

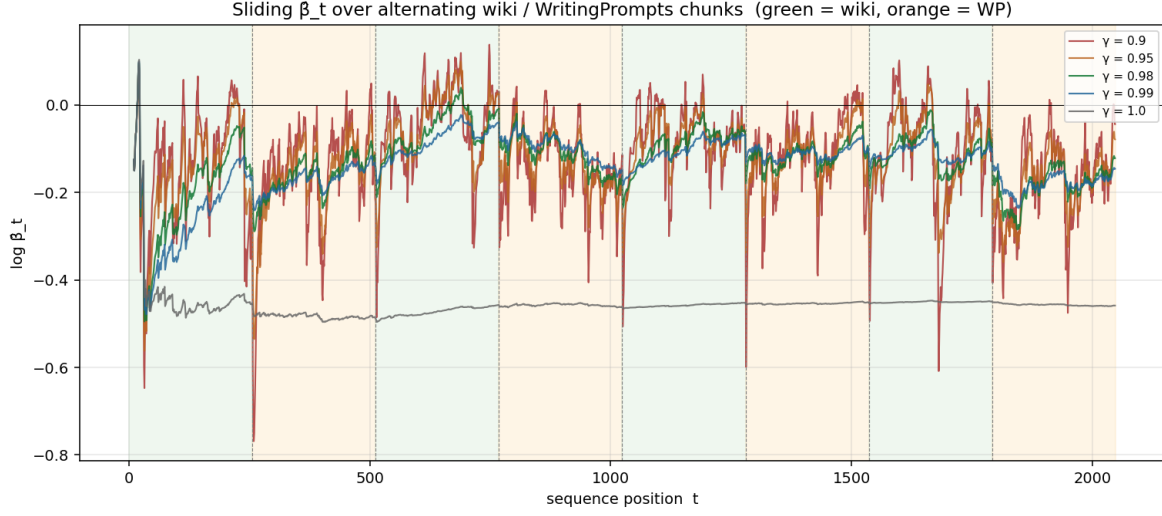


Figure 5: $\log \hat{\beta}_t$ over position; background shading is genre. $\gamma = 1$ (grey) drifts to the pooled bias and stops tracking; damped filters oscillate weakly with genre.

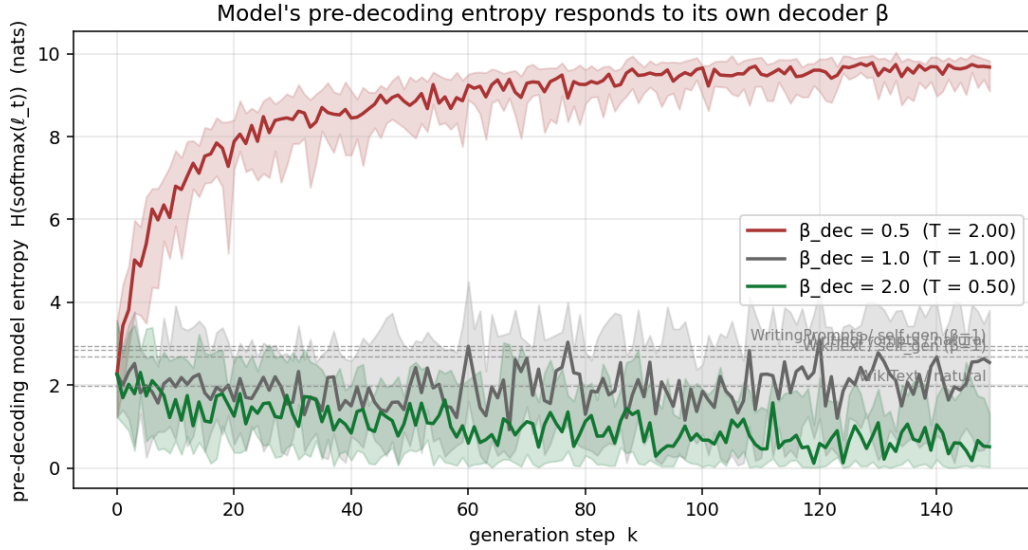


Figure 6: Median pre-decoding $H(t)$ per condition (IQR shaded). Dashed lines mark mean entropies of real natural text; the $\beta = 0.5$ trajectory climbs far past anything natural.

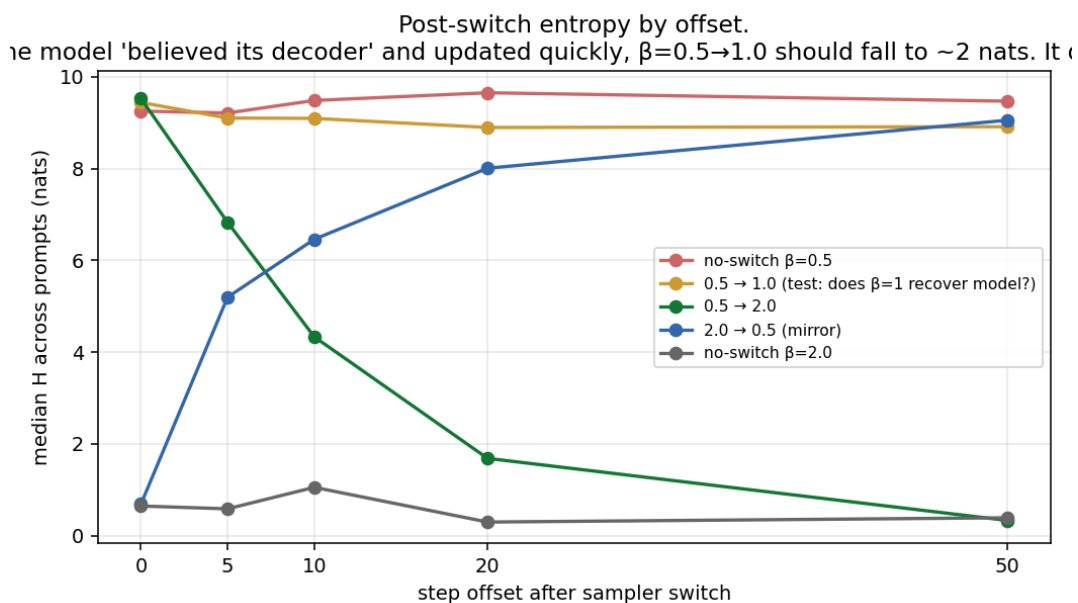


Figure 7: Median pre-decoding H at offsets after the sampler switch. The orange line (pre05_post10, $0.5 \rightarrow 1.0$) does not fall: switching to a neutral sampler does not restore coherence.

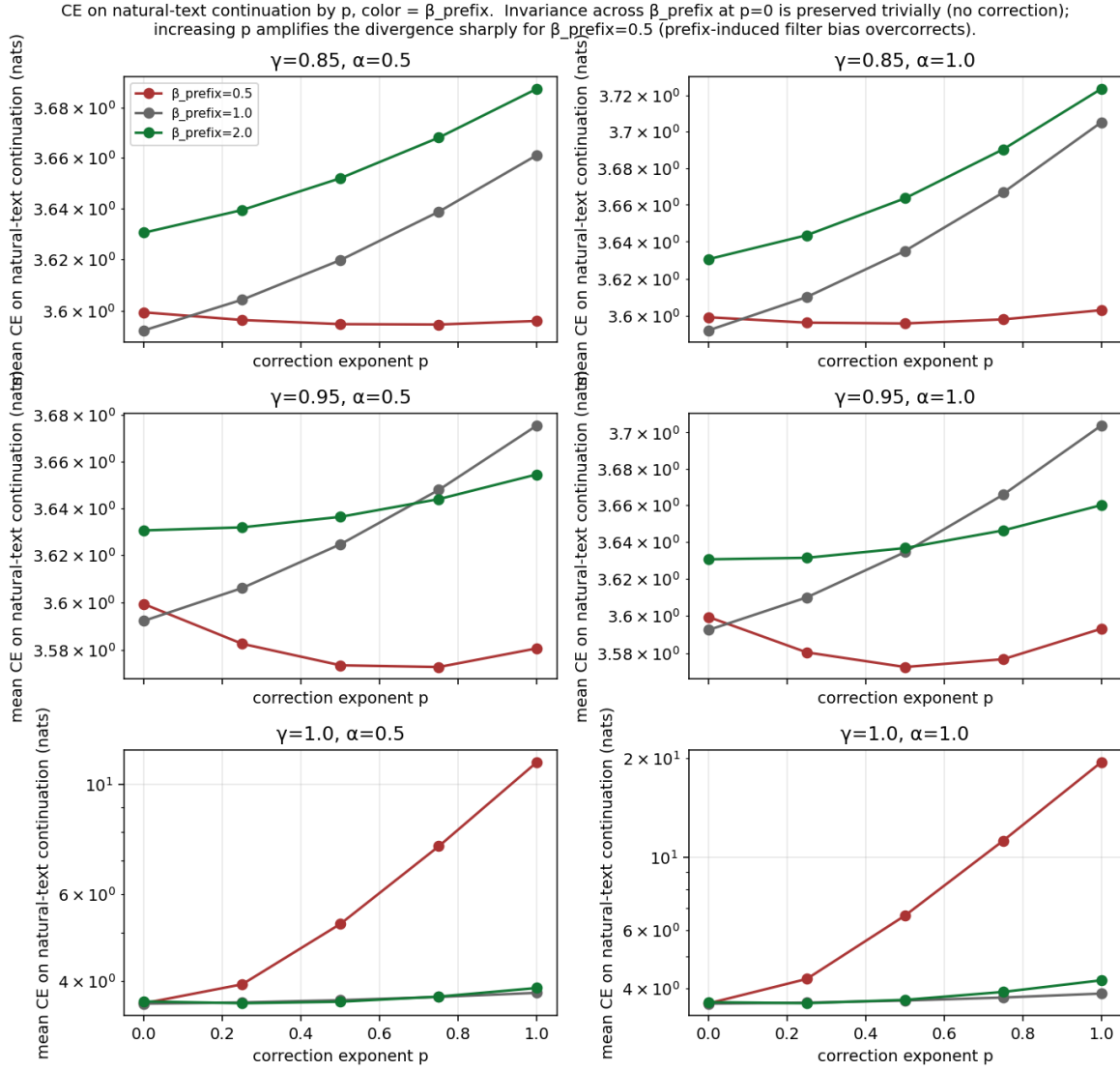


Figure 8: Cross-entropy on the natural-text continuation vs. correction exponent p , colored by β^{prefix} . At $\gamma = 1$ the $\beta = 0.5$ curve grows explosively with p ; at $\gamma < 1$ all curves stay within ≈ 0.1 nat of each other.